St. John College of Engineering and Management
Autonomous Institute
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Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE
DTE Code : 3218 AICTE Permanent ID : 1-4790201
NAAC Accredited with Grade 'A+', Three Programs NBA Accredited

Capstone Project Report

on

Counter-Based Indicator Light Control Using PLC

Submitted in partial fulfillment of the requirements
*of the **Third Year** of*
Bachelor of Technology
In
Electronics and Computer Science

by

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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI
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This is to certify that the following students of B. Tech in Electronics and Computer Science Department have satisfactorily completed the requirements for the courses **Programmable Logic Controllers and SCADA** prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai)**, while working on "Counter-Based Indicator Light Control Using PLC".

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Declaration

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Abstract

This project presents a PLC-based counter control system in which an indicating light is switched ON when a count reaches 5 and switched OFF when the count reaches 10, after which the counter is automatically reset to start a new cycle. The count pulses are assumed to come from a single external source connected to a single input terminal of the PLC.

The system is implemented on a Mitsubishi FX5U PLC and programmed in ladder logic using MELSOFT GX Works3. A single 16-bit data register (D0) acts as the counter: input X0 increments it by one on every pulse using the ADD instruction, a comparison instruction (D0 >= K5) energises the indicator output Y0, and a second comparison (D0 = K10) sets an internal relay M0 that resets D0 to zero through the RST instruction. A manual reset push button (X1) is also provided. Because the output depends on the comparison result, the light stays ON for counts 5 to 9 and turns OFF as soon as the counter is cleared at count 10.

The program was developed and tested using the GX Simulator3 environment. The work demonstrates how counting, comparison and reset logic can be combined to build a simple, reliable and easily modifiable sequence control, which forms the basis of many industrial counting, batching and signalling applications.

CONTENTS

Chapter No.	Topic	Page No.
CHAPTER 1	**OVERVIEW**	**1**
1.1	Introduction	1
1.2	Background	1
1.3	Importance of the Project	1
1.4	Perspective of Stakeholders and Customers	2
1.5	Objectives and Scope of the Project	2
CHAPTER 2	**LITERATURE SURVEY & PROPOSED WORK**	**3**
2.1	Introduction	3
2.2	Literature Survey Table	3
2.3	Problem Definition	4
2.4	Feasibility Study	4
2.5	Methodology Used	4
CHAPTER 3	**ANALYSIS AND PLANNING**	**6**
3.1	Introduction	6
3.2	Project Planning	6
CHAPTER 4	**DESIGN & IMPLEMENTATION**	**7**
4.1	Data Flow Diagram (DFD)	7
4.2	Block Diagram	7
4.3	Flow Chart	9
4.4	UML & PLC Ladder Diagram	9
CHAPTER 5	**RESULTS & DISCUSSION**	**12**
5.1	Actual Results	12
5.2	Future Scope	13
CHAPTER 6	**CONCLUSION**	**14**
	REFERENCES	**14**

CHAPTER 1: OVERVIEW

1.1 Introduction

Programmable Logic Controllers (PLCs) are industrial computers that monitor inputs, execute a stored program and control outputs in real time. Among the most common tasks given to a PLC is counting: the number of bottles passing on a conveyor, the number of products packed in a box, or the number of machine cycles completed. Counting is usually combined with an action, such as turning on a signal lamp or stopping a machine when a preset value is reached.

This project implements one such task. An indicating light is to go ON when a count reaches 5 and go OFF when the count reaches 10. When the light goes OFF, the counter must be reset so that the cycle can repeat. The count pulses come from a single external source connected to a single input terminal of the PLC. The solution is built on a Mitsubishi FX5U PLC using MELSOFT GX Works3 ladder logic and verified using the GX Simulator3.

1.2 Background

Before PLCs, counting and signalling were done with electromechanical counters, relays and timers wired together for every new requirement. Any change in the count value meant rewiring the panel. A PLC replaces this hard wiring with a program: the same hardware can count to 5, 10 or 100 simply by changing a constant in the ladder diagram.

Most PLC platforms offer dedicated counter elements as well as arithmetic and comparison instructions. In this project the count is held in a data register (D0) which is incremented with the ADD instruction on every input pulse. Using a register with comparison instructions makes it easy to produce two different actions at two different count values (ON at 5, OFF at 10) from one single counting element and one single input terminal, without needing two separate counters wired to the same pulse source.

1.3 Importance of the Project

- **Single-Input Counting:** All pulses arrive on one input terminal (X0), which keeps the wiring simple and saves input points on the PLC.
- **Two-Threshold Control:** One counter register drives two different events, light ON at count 5 and light OFF at count 10, using comparison instructions.
- **Automatic Reset:** The counter clears itself when the light goes OFF, so the machine is immediately ready for the next cycle without operator action. A manual reset (X1) is also available.
- **Flexibility:** The preset values (K5 and K10) can be changed in software in seconds. The same logic can be reused for batch counting, packaging and production monitoring.

- **Foundation for Industrial Applications:** The concepts of counting, comparison, latching and resetting used here are the building blocks of larger sequence-control programs.

1.4 Perspective of Stakeholders and Customers

- **Students and Educators:** A compact, complete example of counter logic, comparison instructions and reset techniques in ladder diagram programming.
- **Plant Operators:** A clear visual indication (lamp) tells the operator when a preset quantity has been reached, reducing manual counting errors.
- **Industries (packaging, bottling, assembly):** A low-cost, easily modifiable counting and signalling solution that can be scaled to real production lines.

1.5 Objectives and Scope of the Project

Problem Statement: An indicating light is to go on when a count reaches 5 and to go off when a count reaches 10. Reset counters when the light goes off. Assume the count pulse is coming from a single external source and it is connected to a single input terminal.

Objectives:

1. To study the counting and comparison capabilities of the Mitsubishi FX5U PLC.
2. To design and develop a ladder diagram in MELSOFT GX Works3 that turns an indicating light ON at count 5 and OFF at count 10.
3. To reset the counter automatically when the light goes OFF, and to provide a manual reset input.
4. To verify the program using the GX Simulator3 and document the results.

Scope:

The project covers a single pulse input, one counting register, one indicator output and one reset input. The program is verified in simulation on the FX5U CPU. Hardware wiring of field devices (pulse sensor and lamp) is described in the design chapter, but the project does not include multi-stage counting, HMI or SCADA display, which are listed as future scope.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

Counting and comparison are standard topics in PLC programming. Textbooks, manufacturer manuals and the IEC 61131-3 standard describe the instructions available for counting, arithmetic, comparison and resetting. This chapter reviews the key references used for the project, identifies what they provide and what remains to be designed by the programmer, and defines the problem and methodology.

2.2 Literature Survey Table

Table 1: Literature Survey

Sr. No.	Title of Book / Document	Authors & Year	Technology / Content Used	Limitations / Research Gap
1	Programmable Logic Controllers (6th ed.)	W. Bolton, Newnes / Elsevier, 2015	Ladder programming, internal relays, timers and counters	General teaching text; does not give a ready solution for ON at one count and OFF at another with automatic reset from a single input.
2	MELSEC iQ-F FX5U User's Manual (Hardware) and FX5 Programming Manual	Mitsubishi Electric Corporation	FX5U I/O devices (X, Y, M, D), ADD, RST and comparison instructions	Reference documentation only; instructions must be combined by the programmer to build the required sequence.
3	MELSOFT GX Works3 Operating Manual	Mitsubishi Electric Corporation	Ladder editor, project creation, GX Simulator3 monitoring	Explains the tool; the logic design for this specific problem is not covered.
4	IEC 61131-3: Programmable Controllers, Part 3: Programming Languages	International Electrotechnical Commission, 2013	Standard definition of ladder diagram and counter function blocks (CTU, CTD, CTUD)	Defines the language; counter behaviour at two thresholds and reset timing depend on the programmer's design.
5	Programmable Logic Controllers	F. D. Petruzella, McGraw-Hill Education	Up/down counters, counter reset and comparison examples	Examples use generic PLC notation; an FX5U-specific implementation with a data register counter must be developed.

Sr. No.	Title of Book / Document	Authors & Year	Technology / Content Used	Limitations / Research Gap
6	Programmable Logic Controllers: Principles and Applications (5th ed.)	J. W. Webb and R. A. Reis, Prentice Hall, 2003	Counting and sequencing logic, latching and reset circuits	Does not address a single-input, two-threshold indicator with auto-reset on the FX5U.

Research gap: The references explain how counters, comparisons and resets work, but none of them provides a complete, tested solution for the specific requirement of this project. The proposed work combines these instructions into a compact four-rung program for the FX5U and verifies it in simulation.

2.3 Problem Definition

The requirement is to control an indicating light from a count of pulses arriving on a single input terminal. The following points define the problem:

1. **Single Input, Two Events:** The same pulse input must produce an ON event at count 5 and an OFF event at count 10. Therefore one counter must be compared against two different values.
2. **Light State Between Counts:** The light must remain ON for all counts from 5 up to 9 and must not flicker or turn off before the count reaches 10.
3. **Reset After Light Goes OFF:** The counter must return to zero when the light goes OFF, so that the next cycle starts from 0 without operator intervention.
4. **Manual Override:** An operator must be able to clear the counter at any time with a reset button.

2.4 Feasibility Study

- **Technical Feasibility:** The Mitsubishi FX5U provides 24 V DC digital inputs (X), transistor outputs (Y), internal relays (M) and data registers (D). The ADD, RST and comparison instructions needed for the program are part of the standard instruction set. The program can be simulated in GX Works3 without any hardware.
- **Operational Feasibility:** The program has only four rungs with clear labels, so it is easy to understand, test and modify. The operator only needs to observe one lamp and, if required, press one reset button.
- **Economic Feasibility:** The project needs only a PLC, a pulse source (push button or sensor) and an indicator lamp. Development and testing use free simulation features of the programming software, so the cost is minimal.

2.5 Methodology Used

The system follows a six-stage sequence of operation:

1. **Pulse Input:** The external pulse source is connected to PLC input X0. Every pulse (rising edge) is detected once per occurrence.
2. **Counting (Rung 1):** On each pulse, the instruction ADD K1 D0 D0 increases the count register D0 by one.
3. **Light ON (Rung 3):** The comparison D0 >= K5 becomes true when the count reaches 5 and energises output Y0 (indicating light).
4. **Light Held ON:** The comparison stays true for counts 5, 6, 7, 8 and 9, so the light stays ON.
5. **Auto-Reset Flag (Rung 4):** When Y0 is ON and D0 = K10, internal relay M0 is energised.
6. **Reset (Rung 2):** M0 (or manual reset X1) executes RST D0, clearing the count to 0. The comparison D0 >= K5 becomes false, Y0 turns OFF and M0 drops out. The system is ready for the next cycle.

CHAPTER 3: ANALYSIS AND PLANNING

3.1 Introduction

Successful execution of the project requires structured partitioning of work across requirement analysis, I/O assignment, ladder logic development, simulation and testing, and report preparation. The four-member team divided these activities so that design, programming, testing and documentation could proceed in parallel.

3.2 Project Planning

Table 2: Project Planning & Phase Breakdown

Phase	Activity Description	Deliverable	Duration
Phase 1	Requirement analysis and literature review	Problem statement and instruction selection	Week 1
Phase 2	I/O assignment and logic design (DFD, block diagram, flow chart)	I/O table and design diagrams	Week 2
Phase 3	Ladder logic development in GX Works3	Ladder program (.gx3 file)	Week 3
Phase 4	Simulation and testing in GX Simulator3	Verified program and result screenshots	Week 4
Phase 5	Documentation and report preparation	Final project report	Week 5

CHAPTER 4: DESIGN & IMPLEMENTATION

4.1 Data Flow Diagram (DFD)

Level 0 shows the system as a single process that receives pulses from the external source and a reset command from the operator, and gives an ON/OFF signal to the indicator lamp. Level 1 breaks the process into counting, comparison and reset functions.

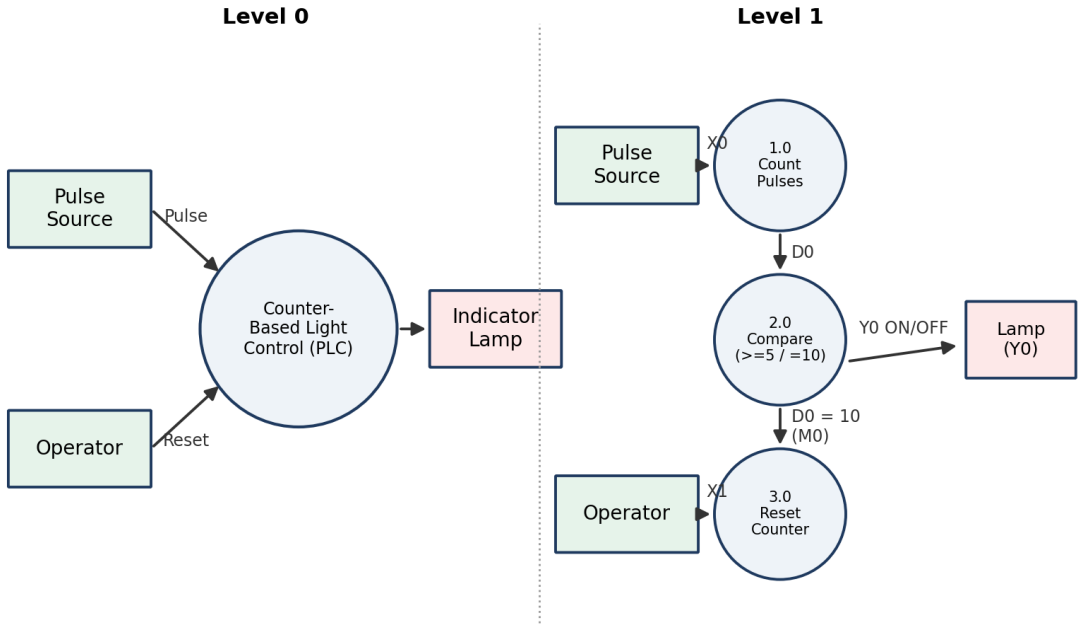


Figure 1: DFD Level 0 & Level 1

4.2 Block Diagram

The count pulse source and the manual reset button are connected to the PLC input terminals X0 and X1. The CPU executes the ladder program every scan and drives the output Y0, which is connected to the indicating light through the output module.

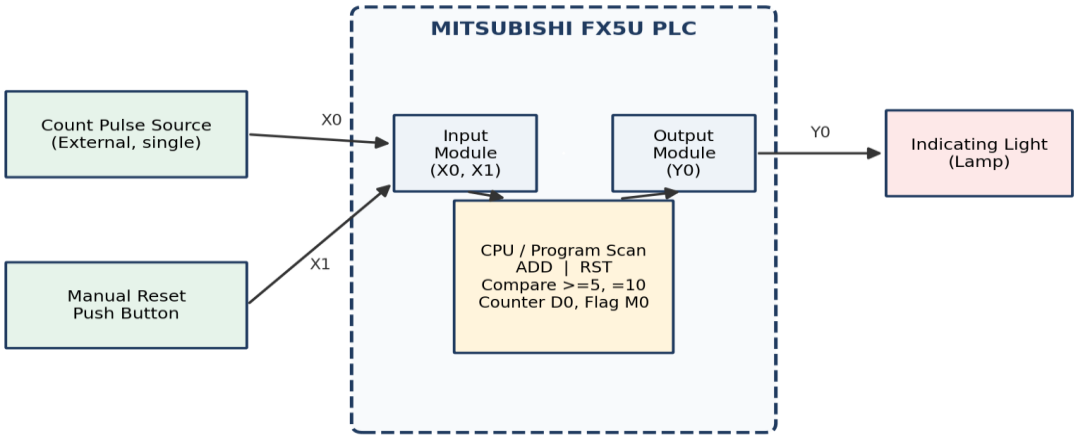


Figure 2: System Hardware & PLC Interfacing Block Diagram

4.3 Flow Chart

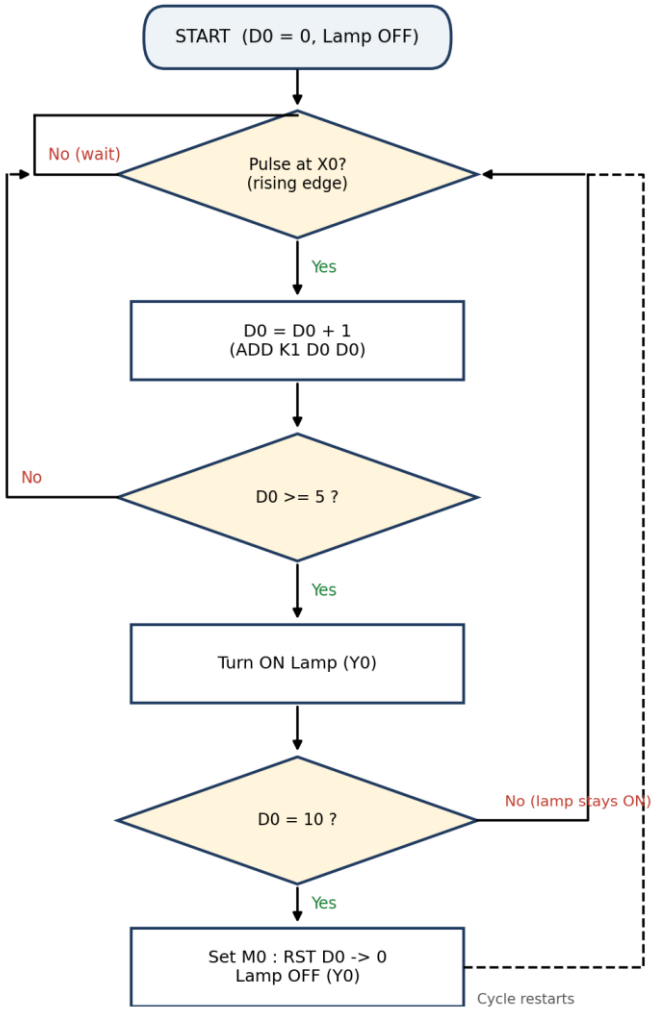


Figure 3: Counting and Indicator Light Control Flow Chart

4.4 UML & PLC Ladder Diagram

The PLC addresses used in the program are listed in Table 3.

Table 3: Mitsubishi FX5U PLC I/O & Internal Device Allocation

Device Tag	Type	Function / Description
X0	Digital Input	Count pulse input from the single external source
X1	Digital Input	Manual reset push button
D0	Data Register	Count value (incremented on every pulse of X0)
M0	Internal Relay	Auto-reset flag, set when the count reaches 10
Y0	Digital Output	Indicating light (ON at count 5, OFF at count 10)

Ladder Diagram Explanation

- **Rung 1 (step 0):** On every rising edge of the pulse input X0, the instruction ADD K1 D0 D0 adds 1 to the count register D0.
- **Rung 2 (step 9):** If the manual reset X1 or the auto-reset flag M0 is ON, the instruction RST D0 clears the counter to zero.
- **Rung 3 (step 16):** The comparison D0 >= K5 energises output Y0. As long as the count is 5 or more, the light is ON.
- **Rung 4 (step 22):** When Y0 is ON and D0 = K10, internal relay M0 is energised. In the next scan, Rung 2 resets D0 to 0, Y0 turns OFF because the count is below 5, and M0 drops out.

Since the program is executed from top to bottom in every scan, the reset triggered by M0 takes effect in the scan following the one in which the count reached 10. This delay is only one scan time (a few milliseconds) and is not noticeable.

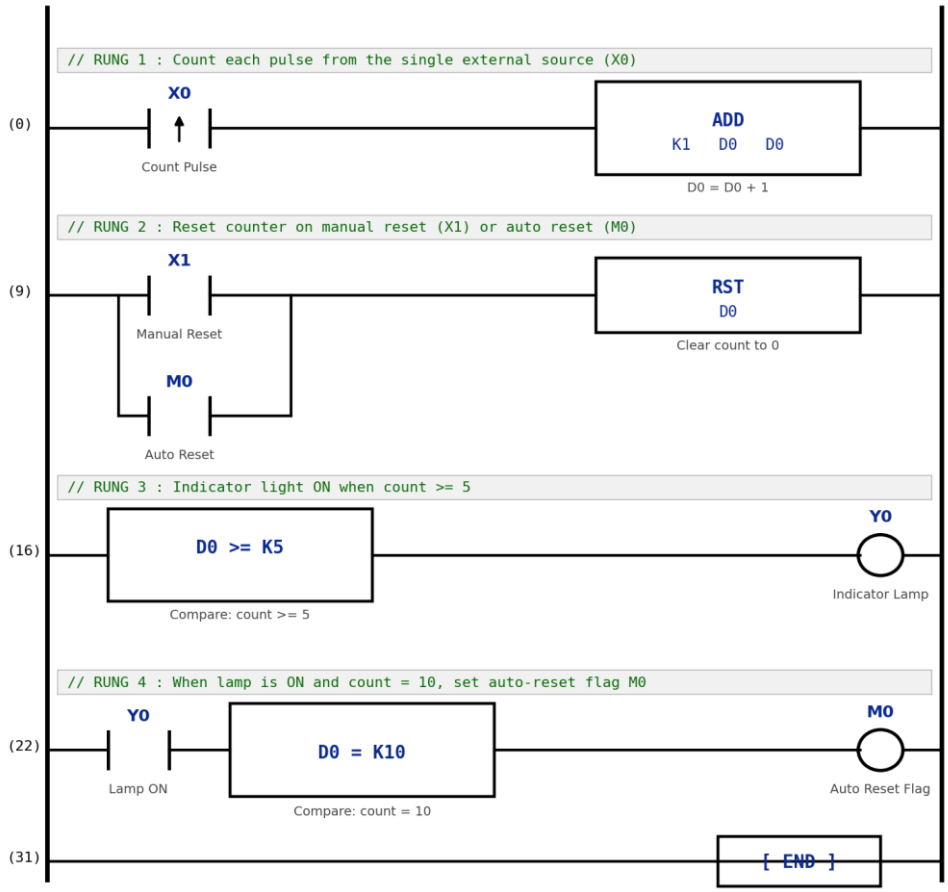


Figure 4: Mitsubishi FX5U PLC Ladder Diagram (MELSOFT GX Works3 Format)

Table 4: Logic Operation Table

Count (D0)	Y0 (Light)	M0 (Reset flag)	Remark
0 – 4	OFF	OFF	Counting in progress; D0 >= K5 is false
5	ON	OFF	Light turns ON at count 5
6 – 9	ON	OFF	Light stays ON as D0 >= K5 remains true
10	ON → OFF	ON (one scan)	M0 resets D0 to 0; light turns OFF
0 (after reset)	OFF	OFF	New cycle starts

The expected timing relationship between the input pulses, the counter value and the indicating light is shown in Figure 5.

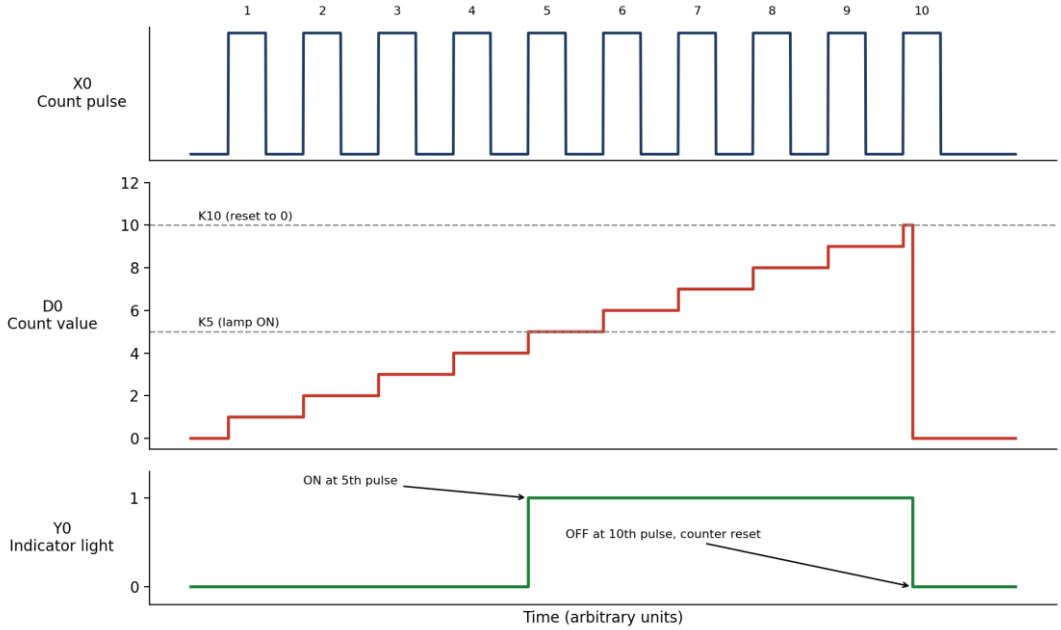


Figure 5: Timing Diagram (Count Pulses, Counter Value D0 and Indicator Y0)

CHAPTER 5: RESULTS & DISCUSSION

5.1 Actual Results

The ladder program was written in MELSOFT GX Works3 and tested in the GX Simulator3 using the FX5UCPU simulation. The simulator is in RUN mode with the PWR and P.RUN indicators ON and the ERR indicator OFF, which confirms that the program was downloaded and is executing without errors.

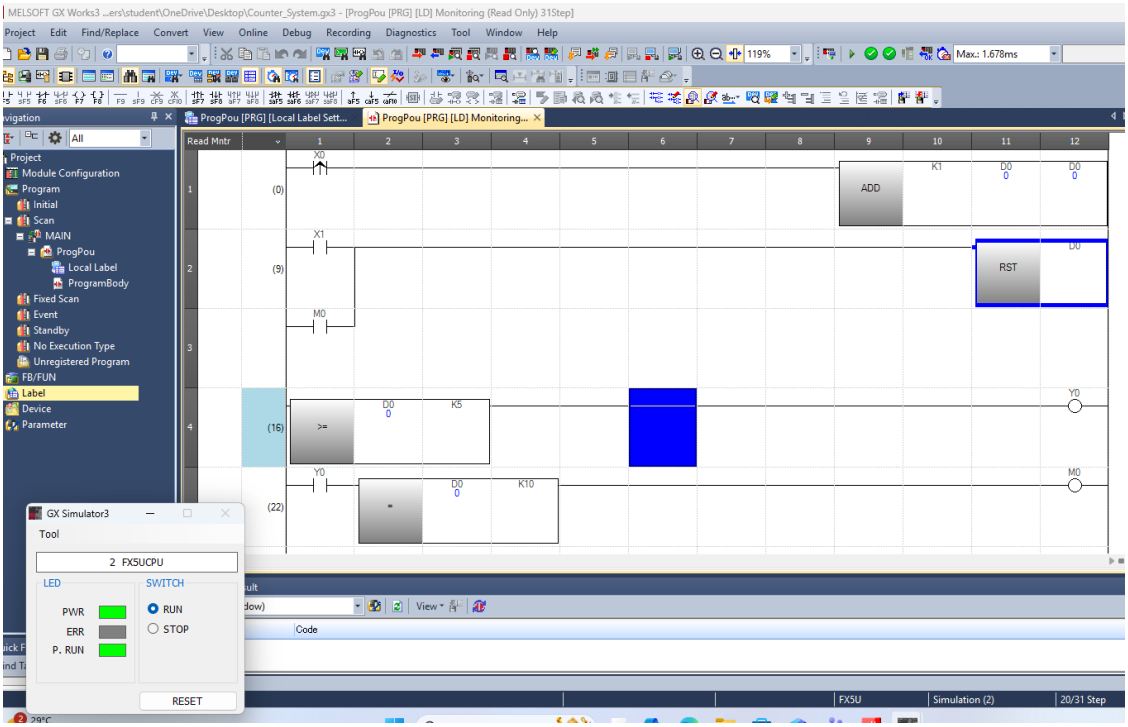


Figure 6: GX Works3 Monitoring Window with GX Simulator3 (FX5UCPU in RUN mode)

The following results were observed from the simulation:

- **Counting Accuracy:** Each pulse on X0 increased the value of D0 by exactly one.
- **Light ON at Count 5:** Output Y0 turned ON when D0 reached 5 and stayed ON for counts 5 to 9.
- **Light OFF at Count 10 and Reset:** When D0 reached 10, M0 was set, D0 was cleared to 0 and Y0 turned OFF.
- **Repeatable Cycle:** After the reset, the next set of pulses repeated the same sequence without any operator action.
- **Manual Reset:** Turning X1 ON cleared the counter to zero at any point and turned the light OFF.

5.2 Future Scope

1. **Adjustable Preset Values:** Store the ON and OFF counts (5 and 10) in data registers so they can be changed from an HMI or touch panel without editing the program.
2. **Real Sensor Input:** Replace the push button with a proximity or photoelectric sensor to count objects on a conveyor, with input filtering for noise and contact bounce.
3. **SCADA / HMI Monitoring:** Display the live count, light status and cycle count on a SCADA screen for remote monitoring and data logging.
4. **Multi-Stage Indication:** Extend the program to several lamps or outputs at different count values, such as a warning lamp, a batch-complete lamp and a buzzer.
5. **Power-Failure Retention:** Use latched (battery-backed) registers so the count is retained after a power loss.

CHAPTER 6: CONCLUSION

The counter-based indicator light control was successfully implemented on a Mitsubishi FX5U PLC using MELSOFT GX Works3 ladder logic. A single data register (D0), incremented by pulses on one input terminal (X0), was compared with the values 5 and 10 to switch the indicating light ON at count 5 and OFF at count 10, and the same event reset the counter automatically for the next cycle. A manual reset input (X1) was also provided. The four-rung program was verified in GX Simulator3 and shows that complex-looking sequence requirements can be met with a few basic PLC instructions. The same approach can be extended to industrial counting, batching and signalling applications.

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